

key logging, which enables it to record anything that the user types, including the user's passwords, credit card number, e-mail messages, chat messages, etc. Some spyware can even spy on the user by exploiting his own webcam.

Scareware. Scareware is a software that creates fear in users by asking them to perform some kind of activity while threatening with the adverse effects of not following the guidelines. Scareware can install malicious software that may steal information, make the system unstable or even crash it.

Other malware. Some other kinds of malware (malicious software) include scamware, drug dealerware and theftware. Scumware is designed to steal traffic and revenue from legitimate websites. Drug dealerware offers free software and then shuts down and demands payments months later when one has presumably become used to it. Theftware hijacks ad-space on webpages, replacing the ads space with its own ads.

Spam

Spam is unsolicited junk e-mail that can take the form of advertisements, chain mail, bulk e-mail, threatening or abusive e-mail, etc. Spammers often use tools like 'harvesters' that scan the Internet and newsgroup and collect e-mail addresses. Spammers may also buy a list of e-mail addresses from a website that holds user information in its database.

Most of the spams are used for fraudulent advertisements like won-lottery-jackpot announcements, get-rich-quick business opportunities, free gifts and work-at-home schemes. These spams may carry viruses or try to lure the user into providing some personal and financial information including user-ID, password, credit card number, etc.

Some dangerous spams come from worms, not spammers. These generate an infected e-mail from the systems of unsuspecting hosts. Some spams are circulated unknowingly through a common user who passes on chain letters, devotional messages, pleas for medical

help, etc. Such type of chain letters or other things promise a large return for small effort and often also threaten bad luck if one breaks the chain.

Denial-of-service attacks

Denial-of-service attack (DoS), as the name suggests, is a kind of cyber attack that makes a system, network resource, network components, website, or services hosted on reputed and important web servers unavailable temporarily or permanently.

Generally, there are two forms of DoS attacks: those that crash services and those that flood services. In its simplest form, DoS sends a large quantity of communication requests to a targeted resource in order to make the resource busy, saturate or overflow so that it becomes unavailable for a certain period of time or responds so slowly as to be rendered essentially unavailable. Such attacks may consume system resources (such as bandwidth, memory and processor time), disrupt configuration information (such as routing information), disrupt physical network component or even force systems to reset so as to make the system unavailable to its intended users.

Moreover, DoS may include execution of malicious software. Attackers frequently use compromised systems to form botnet. These compromised systems (bots) are then used as the launch pad for attacking other systems. This kind of attack is known as distributed denial-of-service (DDoS) attack.

Often intruders install 'agent' on several compromised systems awaiting command from the intruder. A single 'handler' instructs all such compromised systems to launch the attack on another system. If such attack is conducted on a sufficiently large scale, it may cause a serious network security event that may be problematic for users, service providers and law enforcement agencies.

Variants of DoS include smurf attack, ping flood, ping of death, SYN flood, teardrop, spoofed/reflected attack and unintentional attack (Fig. 2).

In smurf attack, a misconfigured

network component allows packets to be sent to all systems on a particular network via the broadcast address of the network.

In ping flood attack, a large number of ping packets are sent to the targeted system, usually using 'ping' command.

In ping-of-death attack, a deformed ping packet is sent to the victim that can crash the system.

In SYN flood, the attacker sends a flood of TCP/SYN packets with a forged source address. In such a case, due to forged source address, the victim's computer keeps waiting for a response from the source address and ultimately remains unavailable to its intended users.

Teardrop attack sends IP fragments with overlapping and over-sized payloads.

Spoofed or reflected attack involves setting of source address to that of the targeted system. This results in a large number of replies from several computers to the targeted system. Sometimes, due to a sudden enormous spike in the popularity of a website, denial-of-service event on such a less-equipped website occurs due to access requests from a huge number of people.

To sum up

Defending against cyber threats typically involves the use of a combination of attack detection, traffic classification, prevention and recovery from a security attack. There are a lot of techniques for defence. Keeping all the application software and operating system updated, changing passwords frequently, using firewall, antivirus and antispyware software, cryptographic tools, intrusion detection system and software/hardware/physical controls, and above all educating people on security solves many problems. ●

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